
h
b
ModernBERT-base
frozen
LoRA
$\mathbf{q}_v$
H
Conflict
Novelty
Prominence
Human interest
$[\mathbf{e}_h; \mathbf{e}_b]$
Impact
Proximity
Timeliness
$\mathcal{L}_{\text{CORAL}}$ vs. y_v
$S(x)$
Σ
C
π
1
2
3
4
5
no clicks or user signals
training only
F
interaction
Σ_w
z
$c(x)$
list-wide head
$\mathcal{L}_{\text{ListNet}} + \mathcal{L}_{\text{ApproxNDCG}}$ vs. y_{gold}
$\mathcal{L}_{\text{wide}}$